# Advanced Data Engineering 02: Data Categorization and Summarization


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# Agenda

- Data categorization:
  - based on data structure
  - based on statistical/ measurement scales
  - based on content and context
- Data summarization:
  - descriptive statistical summaries
  - graphical methods
  - tabular summaries
  - data aggregation techniques


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# Data categorization

- Understanding data categories is crucial for data governance, database design, data integration, and analytics. They help determine:
  - How data is stored and retrieved
  - The methods used for data processing and analysis
  - The appropriate data management strategies and tools
- Data categorization:
  - based on data structure
  - based on statistical/ measurement scales
  - based on content and context

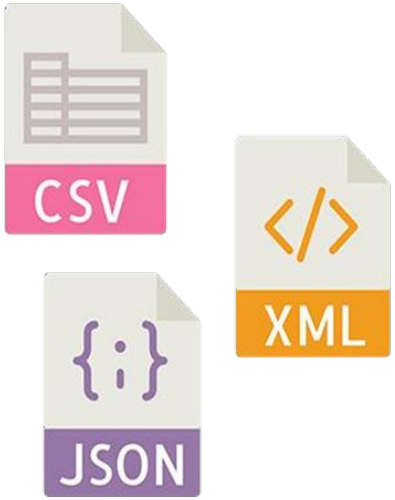

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# Data categorization based on data structure

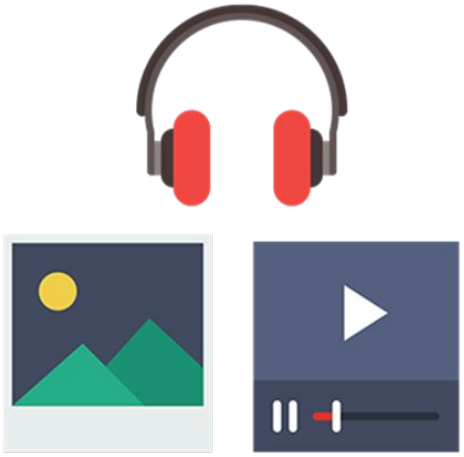
**Structured Data**



**Semi-Structured Data**




**Unstructured Data**




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# Feature vs. Data Points & Data Dimensionality

- Speaking of data, especially in tables, we refer to **features**, sometimes also called variables/ attributes, and **data points**.
- Features are the distinct attributes or properties of the data set. In tabular data, these often appear as columns.
- Data points are individual pieces of information, often represented as rows in tabular data.
- The concept of **dimensionality** arises from the **number of features**.


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# Based on statistical/ measurement scales

- **Categorical (Qualitative)** data: can be grouped based on qualitative properties.
  - **Nominal** data: Labels or names with no inherent order (e.g., gender, colors).
  - **Ordinal** data: Categories with a meaningful order but without consistent intervals (e.g., rankings like “low,” “medium,” and “high”).
- **Numerical (Quantitative)** data: represents measurable quantities
  - **Interval** data: Numerical data with meaningful intervals but no true zero point (e.g., temperature in Celsius).
  - **Ratio** data: Numerical data with a meaningful zero point, allowing for the calculation of ratios (e.g., height, weight, or income).


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# Based on Content and Context

- **Metadata vs. Content Data**
  - **Metadata**: describes other data (e.g., file size, creation date, author).
  - **Content** data: actual data content, such as the text in a document or the pixels in an image.
- **Master, Transactional, and Analytical Data**
  - **Master** data: Core data that is essential for operations (e.g., customer or product information)..
  - **Transactional** data: generated from day-to-day operations (e.g., sales transactions, invoices).
  - **Analytical** data: used primarily for analysis and decision-making, often aggregated or derived from transactional data.


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# Data summarization

- Condensing large or complex datasets into a more understandable form, highlighting key characteristics and patterns.
  - Descriptive Statistical Summaries: Measures of Central Tendency, Variability, Percentiles and Quartiles
  - Graphical Methods: Histograms, Box Plots (Box-and-Whisker Plots), Scatter Plots, Bar Charts and Pie Charts.
  - Tabular Summaries: Frequency and Pivot Tables
  - Data Aggregation Techniques


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# Descriptive Statistical Summaries

- Measures of Central Tendency:
  - **Mean**, **Median**, **Mode** : These values indicate the “center” of the data and give a quick snapshot of typical values.
- Measures of Variability:
  - **Range**, **Variance**, **Standard Deviation**, **Interquartile Range** (IQR): These metrics explain how spread out the data is, which is crucial for understanding data consistency and potential outliers.
- Percentiles and Quartiles:
  - These help in determining the distribution of data by indicating the value below which a certain percentage of the data falls.


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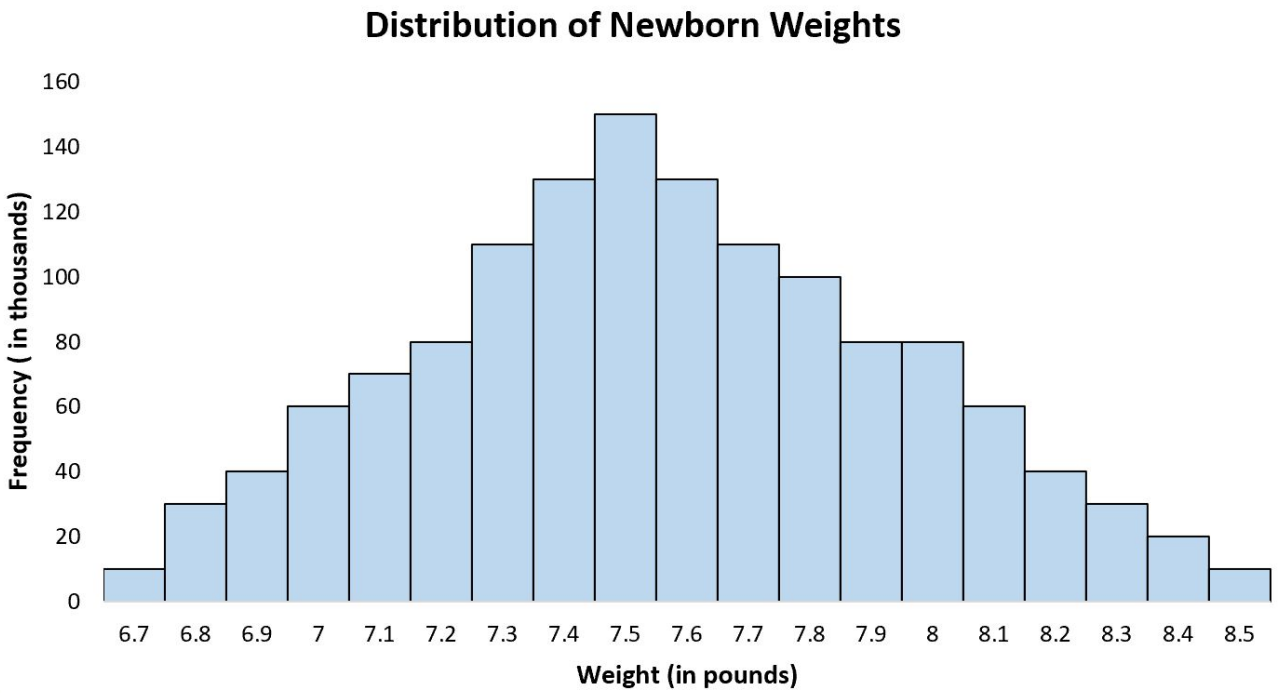
# Descriptive Statistical Summaries

| Scale type | Characteristics | Data type | Operations | Examples |
|---|---|---|---|---|
| Interval | Equal intervals, no true zero | Numerical | Arithmetics, Mean, Median, Standard Deviation | Temperature (Celsius, Fahrenheit) |
| Ratio | Equal intervals, true zero, meaningful ratios | Numerical | Arithmetics, Mean, Median, Standard Deviation | Height, Weight, Age, Income |
| Ordinal | Ordered categories, unequal intervals | Categorical | Median, Percentiles | Ratings (poor, fair, good) |
| Nominal | Categories without order | Categorical | Mode, Percentiles | Colors, Gender |


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# Histogram

- Provide a visual representation of the frequency distribution of numerical data, showing how data points are distributed across different intervals.

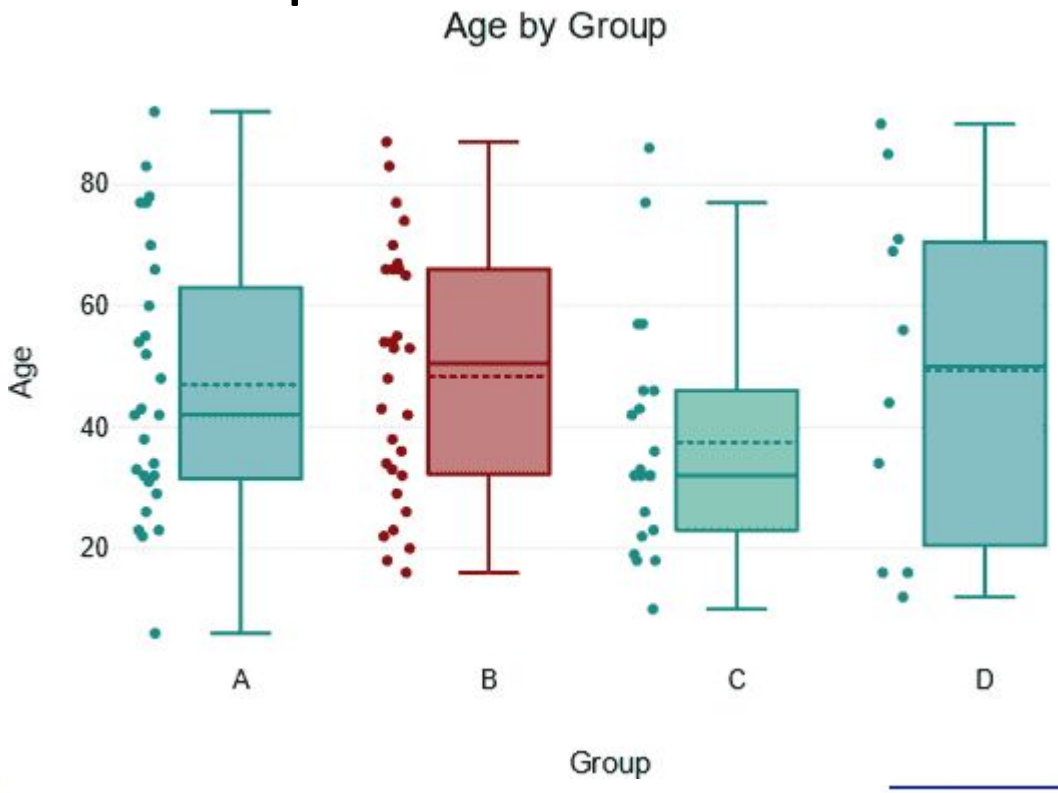




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# Box Plots

- Summarize data by displaying the median, quartiles, and potential outliers, offering a clear view of data dispersion and skewness.

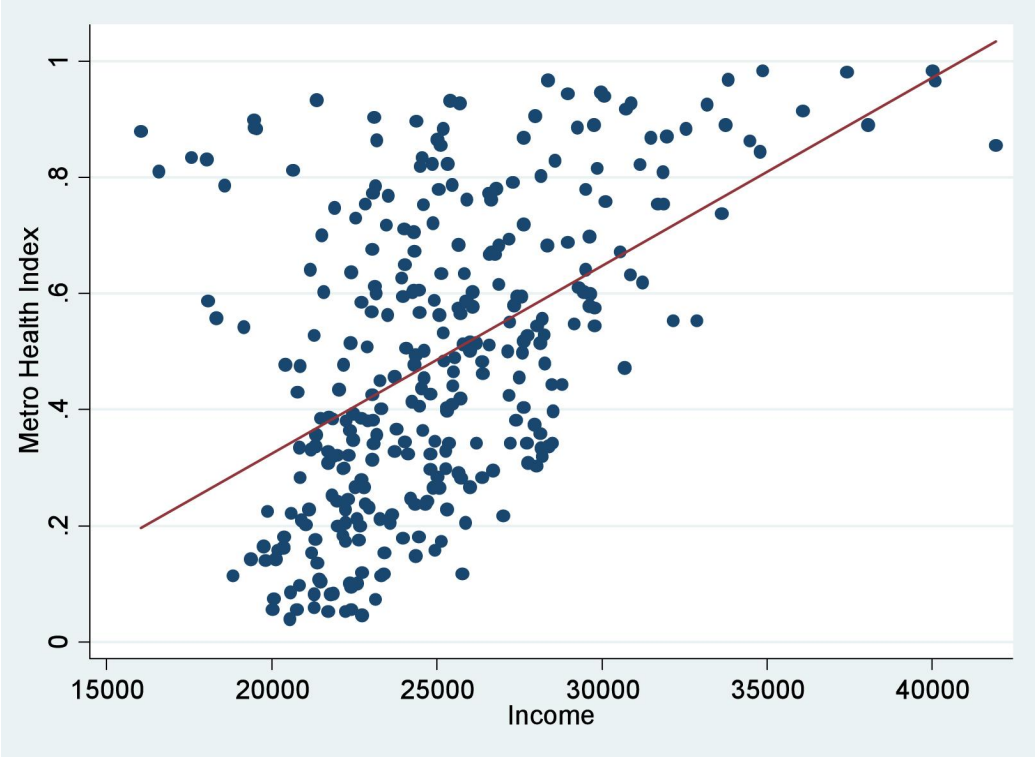




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# Scatter Plots

- Useful for visualizing the relationship between two numerical variables, often revealing correlations or trends.

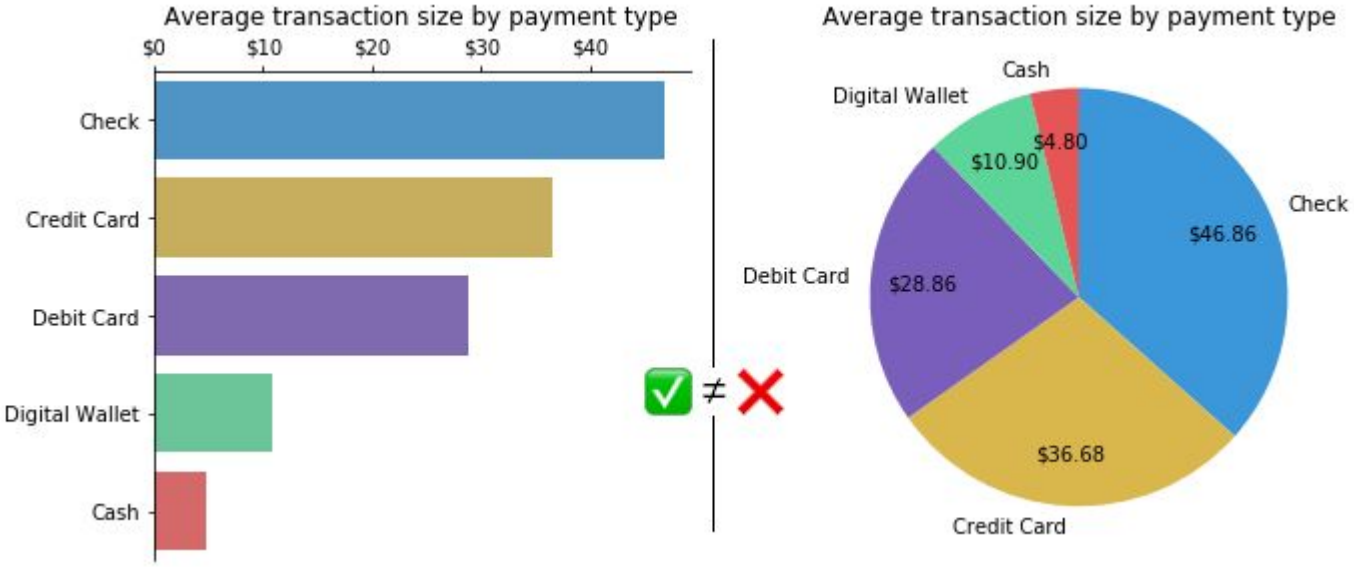




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# Bar and Pie Charts

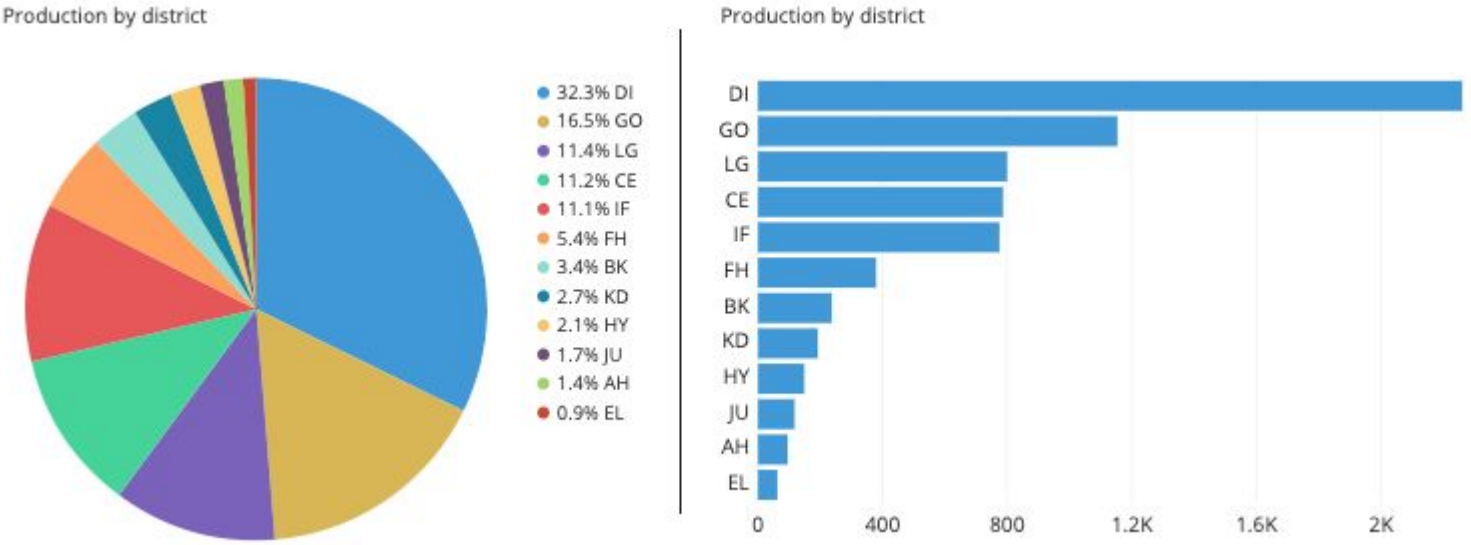
- Ideal for categorical data, these charts illustrate the frequency or proportion of different categories.





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# How to choose bar or pie chart?





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# Tabular summaries

- **Frequency Tables:** List the number of occurrences of each value or category, providing a straightforward summary of data distribution.
- **Pivot Tables**: Allow data aggregation across multiple dimensions (such as summarizing sales by region and product) and are particularly powerful in spreadsheet software.


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# Data Aggregation

- **Grouping Operations:** In databases and programming, “group by” operations help aggregate data by specific categories (e.g., calculating the average sales per region).
- **Time Series Summaries**: For temporal data, methods like moving averages, cumulative sums, or seasonal decomposition summarize trends and patterns over time.


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